

nAD820-13d

8-bit 20 MSPS Analog-to-Digital Converter IP

FEATURES

- TSMC CL013G 1.2/3.3 V Technology
- 8-bit Algorithmic General Purpose ADC
- 100 kSPS to 20 MSPS Conversion Rate
- Excellent Dynamic Performance
47 dBFS SNR at $F_{IN} = 10$ kHz
- Low Power Consumption
9 mW at 20 MSPS
- Power Saving Idle Modes
- 0.25 mm² Core Area
- Pure Internal Voltage Reference
- Single Ended Input Buffer
- Excellent Gain and Phase Matching
- Input Clock Duty Cycle Restorer

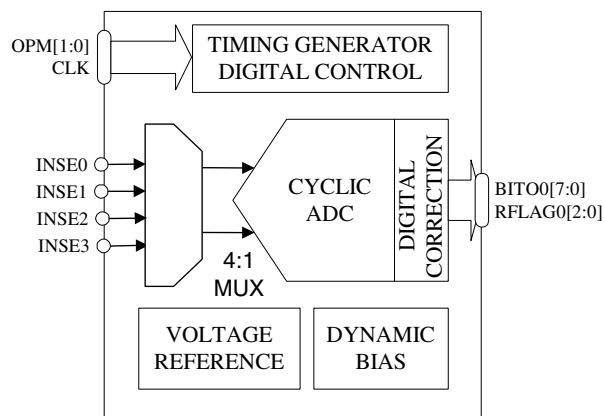


Figure 1. Functional block diagram

APPLICATIONS

- Communication System Control Functions
WLAN 802.11x / WiMAX 802.16x
- Instrumentation

GENERAL DESCRIPTION

The nAD820-13d is a monolithic, high-speed, low power, general-purpose analog-to-digital converter silicon IP. It uses a fully differential algorithmic architecture with digital error correction. The core includes 4:1 input Mux, a single ended input buffer and pure internal voltage reference.

The ADC consumes only 4 mW at 20 MSPS operation. Combined with power saving idle modes the ADC is suitable for battery powered devices. Output data is available in a binary offset coded format. Three out-of-range indicator bits are also available for determining if the input signal is over-range, under-range or out-of-range.

Implemented in a generic 0.13 μ m CMOS process, employing a fully differential architecture and no required external reference decoupling it represents an ideal ADC for highly integrated mixed-signal systems.

QUICK REFERENCE DATA

IP Type / Technology	Hard Macro / TSMC CL013G 1.2/3.3 V 1P6M			
IP Area / Dimensions	0.25 mm ² / 0.62 × 0.4 mm			
Parameter	Min.	Typ.	Max.	Unit
Supply Voltage, Analog	2.7	3.3	3.6	V
Supply Voltage, Digital	1.1	1.2	1.3	V
Power Dissipation, at 20 MSPS		9		mW
Differential Non Linearity		± 0.6		LSB
Integral Non Linearity		± 1.0		LSB
Effective Number of bits, 20 MSPS $F_{IN} = 10$ kHz	7.2	7.9		Bit

Table 1. nAD820-13d quick reference data

**ABOUT NORDIC SEMICONDUCTOR**

Founded in 1983, Nordic Semiconductor ASA (OSE:NOD) is a Fabless Semiconductor specializing in short-range radio communication and high-speed data conversion. Nordic is headquartered in Trondheim, Norway,

MIXED SIGNAL IP OFFERINGS

Leading and emerging Fabless Semiconductor companies around the globe leverage Nordic's portfolio of off-the-shelf mixed signal intellectual property for the design of highly integrated circuits for broadband communication, digital imaging and video.

Our IP portfolio includes medium resolution (8 to 12-bit), high-speed (40 MSPS+) data converters (ADC and DAC) and low jitter clock generation (PLL) implemented in the latest deep submicron CMOS technologies. Featuring low power consumption, low-silicon area, and a minimum of required external components and technology options our products offers cost efficient integration of high-performance data converters in cost sensitive, high volume integrated circuit designs.

DELIVERABLES

Our mixed signal intellectual property is delivered as technology specific hard macros. Upon licensing we provide a complete design-kit and documentation that enables efficient, low risk, design-in and integration. The design kit for our off-the-shelf IPs includes:

- Full datasheet
- Evaluation board and samples
- Silicon validation report
- Integration Guidelines
- Physical design database (flat gds2)
- LVS Netlist (spice compatible)
- Footprint (.lef format)
- HDL model (verilog model)
- Timing model (.lib)
- Design-in and integration support

FURTHER INFORMATION

For further information about this product, such as full datasheet, silicon validation report, availability and licensing terms please contact us at: datacon@nordicsemi.no

DOCUMENT INFO

Brief Product Specification:	nAD820-13d
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